AImoclips: A Benchmark for Evaluating Emotion Conveyance in Text-to-Music Generation

Anonymous Authors

Abstract

Recent advances in text-to-music (TTM) generation have enabled controllable and expressive music creation using natural language prompts. However, the emotional fidelity of TTM systems remains largely underexplored compared to human preference or text alignment. In this study, we introduce AImoclips, a benchmark for evaluating how well TTM systems convey intended emotions to human listeners, covering both open-source and commercial models. We selected 12 emotion intents spanning four quadrants of the valence-arousal space, and used six state-of-the-art TTM systems to generate over 1,000 music clips. A total of 111 participants rated the perceived valence and arousal of each clip on a 9-point Likert scale. Our results show that commercial systems tend to produce music perceived as more pleasant than intended, while open-source systems tend to perform the opposite. Emotions are more accurately conveyed under high-arousal conditions across all models. Additionally, all systems exhibit a bias toward emotional neutrality, highlighting a key limitation in affective controllability. This benchmark offers valuable insights into model-specific emotion rendering characteristics and supports future development of emotionally aligned TTM systems.

Keywords

Text-to-Music Generation, Emotion Conveyance, Human Evaluation, Valence–Arousal Model

1. Introduction

Recent advances in generative AI have transformed creative domains, including music. AI music generation has evolved from unconditional [1, 2, 3] to conditional generation [4, 5, 6, 7] for better human controllability, with text-to-music (TTM) systems becoming prominent due to their familiar input format, user-friendliness, and expressive potential. Studies on the evaluation of TTM systems have also followed recent advances [8, 9, 10], focusing on human preferences and text prompt alignment.

However, emotional aspects—central to music perception—are still often overlooked in evaluations of TTM systems. Although several studies have addressed emotion-conditioned music generation, these have mainly focused on symbolic music [11, 12, 13, 14, 15], and the question of how well TTM systems convey intended emotions to human listeners remains largely unexamined. Neglecting emotion evaluation can lead to mismatches between user expectations and generated music, limiting the use of TTM systems in real-world applications such as affective music recommendations or adaptive soundtracks. Thus, assessing the effectiveness of emotional communication is crucial for developing user-centered AI music systems.

In light of this background, we develop AImoclips, a benchmark that consists of AI-generated music clips for evaluating emotion conveyance of TTM systems. We generated 991 songs with six state-of-the-art models, and collected valence–arousal ratings from 111 human participants. We open-source the dataset annotated with valence and arousal scores averaged by multiple raters. We also analyze the results to understand the characteristics of TTM systems regarding emotion conveyance, focusing on three main questions: (1) How differently do TTM systems convey intended emotions to humans? (2) What are the valence–arousal features of well- and poorly conveyed emotions? (3) How emotionally diverse are the songs generated by TTM systems?

2. Related Work

Music emotion analysis has led to the development of several publicly available datasets with emotion annotations. These include categorical [11, 16, 17, 18] and dimensional approaches [12, 19, 20, 21], often

HCMIR25: 3rd Workshop on Human-Centric Music Information Research, September 20, 2025, Daejeon, Korea

using valence–arousal models. Recent benchmarks such as GlobalMood [22] address the cross-cultural differences in music emotion recognition.

In parallel with these developments, evaluating emotional fidelity in AI music generation has gained importance. While many studies rely on automatic metrics due to the cost of human evaluation, recent works have begun incorporating human preferences. However, most research in this area still focuses on assessing automatic metrics [8] or improving them to better align with human judgments [9, 10].

Although relatively few studies directly address emotional conveyance, there have been efforts to analyze emotion conveyance in AI-generated music. The work most similar to ours is that of Gao et al. [23], who evaluated emotion conveyance accuracy in the valence–arousal plane using discrete emotion labels. In contrast, our study uses continuous valence–arousal ratings, involves significantly more clips and subjects, and includes a broader set of systems—both open-source and commercial—providing a more comprehensive evaluation of current TTM systems.

3. Dataset Construction

3.1. Emotion Intents

First, we selected 12 emotion words to represent emotional intents when generating music clips. We used two sets of words as candidates. The first set consists of 28 sample words that Russell [24] used to represent the domain of affect. The second set comprises adjectives from the 12-Point Affect Circumplex Model (12-PAC) [25], which do not correspond exclusively to either the valence or arousal axis. From the candidate words, we selected those included in the Warriner et al.'s extension [26] of Affective Norms for English Words (ANEW) [27] and filtered out words for which either the valence or arousal score—ranging from 1.0 to 9.0—fell between 4.0 and 6.0. This step was intended to exclude words with intermediate valence or arousal values, thereby clearly separating the groups of emotion intents corresponding to each quadrant of the valence-arousal plane. Finally, from the remaining words, we selected three for each quadrant, using 5.0 (both for valence and arousal) as the boundary. The selected words are listed in Table 1.

Table 1
Words chosen for emotion intent of TTM generation

Quadrant	Valence	Arousal	Words
Q1	High	High	happy, excited, energetic
Q2	Low	High	angry, anxious, scared
Q3	Low	Low	sad, gloomy, dull
Q4	High	Low	relaxed, calm, tranquil

3.2. TTM System Selection and Clip Generation

Next, we selected six TTM systems, then generated 14 clips for each emotion intent. There are 12 emotion intents, resulting in 168 songs created by each system, forming a dataset of 1,008 AI-generated music clips. All clips were 10 seconds long. Details about textual prompts used for generation and postprocessing of generated clips are explained in Appendix A.

We utilized four open-source models—AudioLDM 2 [28], MusicGen [29], Mustango [30], and Stable Audio Open [31]—and two commercial models, Suno v4.5 [32] and Udio v1.5 Allegro [33]. AudioLDM 2 and Stable Audio Open are diffusion-based generative audio models [34], while MusicGen consists of an autoregressive transformer-based decoder [35], and Mustango is a hybrid model based on both FLAN-T5 [36], an instruction-tuned language model, and a diffusion model. Detailed information on each model we used is summarized in Table 2.

3.3. Human Evaluation

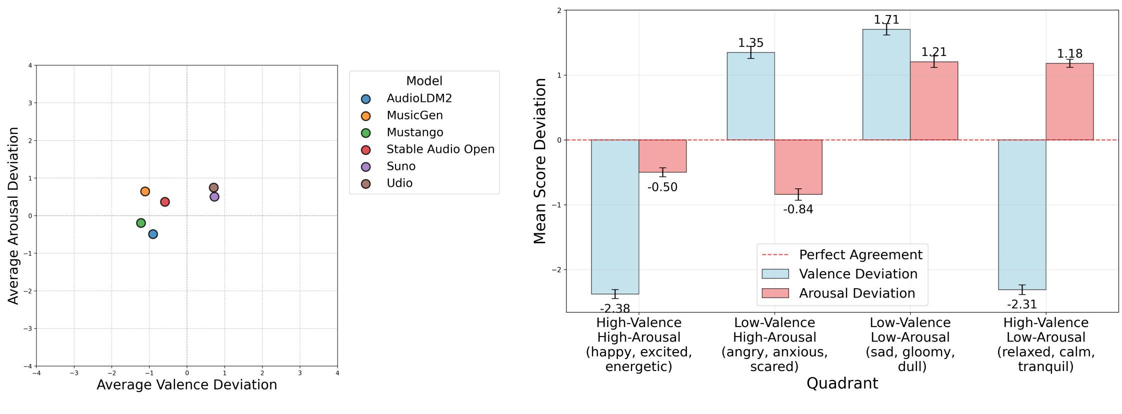
We conducted an online survey to collect valence–arousal ratings for the generated music clips. For each of 111 participants, 56 clips were randomly selected to ensure an average of over five ratings per clip. Each clip was selected from a distinct combination of TTM system and emotion intent, and presented in random order. Participants rated the conveyed valence and arousal on a 9-point Likert scale. All evaluation activites involving human subjects in this study were approved by the IRB at Seoul National University under approval number IRB No.2505/001-012.

4. Results

4.1. Overall Score Distribution for Each TTM System

As shown in Figure 1a, valence and arousal deviations from the intended emotion differ across models. We used scores from Warriner et al's dictionary [26] as the ground truth. Open-source models tend to produce music perceived as less pleasant than intended, resulting in lower valence ratings compared to the ground truth. In contrast, generations from Suno and Udio yield higher valence ratings, forming a clear separation between open-source and commercial models. Arousal deviations are less distinct but still form two clusters: AudioLDM 2 and Mustango show negative arousal deviations, while the others show positive. A two-way ANOVA using TTM system and emotion intent as factors confirms significant model effect for both valence($F_{(5,974)} = 150.53, p < .001$) and arousal($F_{(5,974)} = 39.44, p < .001$) deviance. Moreover, pairwise comparisons (Table 3, Figure 6) support these clusters, showing significant differences between but not within clusters.

Figure 2 and Figure 7 show the distribution of valence-arousal ratings for clips generated by each model. For all models, the clips tend to be rated more neutrally than their original emotion words in text (Ground truth scores are displayed in Figure 7). Ratings for clips generated by Suno and Udio are biased toward higher valence, indicating that these models tend to generate music conveying more pleasant emotions. In contrast, the rating distributions for open-source models tend to lean toward lower valence, with AudioLDM2 and Mustango also exhibiting a bias toward lower arousal. Another notable observation is that Suno and Udio display vertically wider distributions, suggesting that they generate music clips covering a broader range of arousal.



(a) (b)

Figure 1: (a) Mean valence and arousal deviations per model, computed by averaging (clip ratings - corresponding emotion intent scores). (b) Mean deviations per valence–arousal quadrant, aggregated across all models.

4.2. Quadrants in the Valence-Arousal Plane

To analyze which emotions are well or poorly conveyed in the valence-arousal plane, we divided emotion intents into four quadrants and compared how closely the ratings of clips in each quadrant matched the ground truth scores. Figure 1b shows that valence is most accurately reflected when the intended emotion has low valence and high arousal—such as *angry, anxious* and *scared.* For arousal,

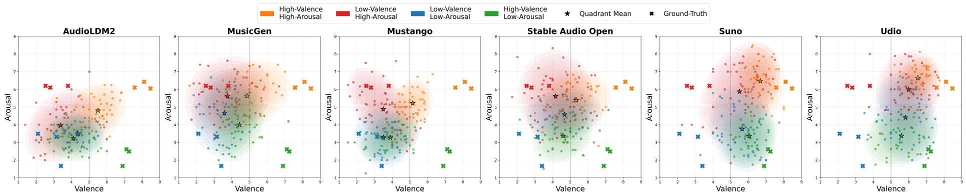


Figure 2: Valence–arousal quadrant distributions by model. Stars show mean ratings per quadrant; ellipses indicate 95% confidence regions.

emotion intents with both high valence and high arousal—such as *happy, excited* and *energetic*—are best conveyed. A two-way ANOVA with the quadrant and TTM system as factors confirms a significant main effect of the quadrant on absolute deviations for both valence($F_{(3,982)} = 29.27, p < .001$) and arousal($F_{(3,982)} = 14.62, p < .001$). In addition, the quadrants with the smallest magnitude of deviations show statistically significant differences from the others (Figure 8, Table 4). The differences between the emotion word scores and the corresponding clip ratings are larger for valence than for arousal, suggesting that the models capture intended arousal more successfully than intended valence.

Looking at the confidence regions displayed in Figure 2, it is notable that the distributions of quadrants with different arousal levels are well separated, whereas those with different valence levels show substantial overlap. In particular, when focusing on the two regions with low arousal level (green and blue regions), valence is not clearly distinguished when arousal level is low. Additionally, for Suno and Udio, the region corresponding to clips generated from high-valence, high-arousal intents is largely overlapped by the region for clips generated from low-valence, high-arousal intents.

5. Discussion

AImoclips is a large-scale benchmark with continuous emotion annotations for evaluating emotional fidelity in TTM generation, incorporating both open-source and commercial systems. By collecting valence–arousal ratings from human listeners and analyzing them statistically, we provide insights into model-specific biases and the current limitations in affective controllability.

Our findings highlight how differently current TTM systems convey emotional intent. Commercial models like Suno and Udio tend to generate music perceived as more pleasant and emotionally engaging than intended, while open-source models often produce less pleasant outputs. One plausible explanation is that listeners perceive more pleasant emotions in music they prefer or in higher-quality audio—Suno and Udio generally receive higher preference ratings [8] and feature higher sample rates than open models (Table 2).

In terms of emotion types, high-arousal emotions such as *excited* and *angry* are more reliably expressed than low-arousal one like *calm* or *gloomy*. This partially aligns with Gao et al. [23], who noted that AI-generated music is particularly effective at conveying high-valence, high-arousal emotions. Additionally, all systems exhibit a centralizing tendency toward emotional neutrality, implying a limited ability to express subtle or polarized affective states. Notably, commercial models show greater arousal variation than valence. They often feature clearer rhythms compared to open models, suggesting that rhythmic entrainment [37] may influence arousal perception.

Future research should investigate the factors that lead to biased valence and arousal perception in music. In addition, the focus should be on figuring out the key traits of TTM systems and music perception for better interactions between those systems and humans in emotion conveyance.

Declaration on Generative AI

During the preparation of this work, the authors used GPT-4.5 in order to: Grammar and spelling check. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

References

[1] A. van den Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, A. Graves, N. Kalchbrenner, A. Senior, K. Kavukcuoglu, Wavenet: A generative model for raw audio, CoRR abs/1609.03499 (2016). URL: http://arxiv.org/abs/1609.03499, arXiv preprint.

[2] C.-Z. A. Huang, A. Vaswani, J. Uszkoreit, N. Shazeer, I. Simon, C. Hawthorne, A. M. Dai, M. D. Hoffman, M. Dinculescu, D. Eck, Music transformer, arXiv preprint arXiv:1809.04281 (2018).

[3] Y.-S. Huang, Y.-H. Yang, Pop Music Transformer: Beat-based modeling and generation of expressive pop piano compositions, in: Proceedings of the 28th ACM International Conference on Multimedia, 2020, pp. 1180–1188.

[4] A. Roberts, J. Engel, C. Raffel, C. Hawthorne, D. Eck, Hierarchical latent vector models for learning long-term structure in music, in: International Conference on Machine Learning (ICML), PMLR, 2018.

[5] C. Payne, Musenet, https://openai.com/blog/musenet, 2019. URL: https://openai.com/blog/musenet, openAI blog post.

[6] P. Dhariwal, H. Jun, C. Payne, J. W. Kim, A. Radford, I. Sutskever, Jukebox: A generative model for music, arXiv preprint arXiv:2005.00341 (2020).

[7] A. Agostinelli, T. I. Denk, Z. Borsos, J. Engel, M. Verzetti, A. Caillon, Q. Huang, A. Jansen, A. Roberts, M. Tagliasacchi, et al., Musiclm: Generating music from text, arXiv preprint arXiv:2301.11325 (2023).

[8] F. Grötschla, A. Solak, L. A. Lanzendörfer, R. Wattenhofer, Benchmarking music generation models and metrics via human preference studies, in: ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2025, pp. 1–5.

[9] Musiceval: A generative music dataset with expert ratings for automatic text-to-music evaluation, in: ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2025, pp. 1–5.

[10] Y. Huang, Z. Novack, K. Saito, J. Shi, S. Watanabe, Y. Mitsufuji, J. Thickstun, C. Donahue, Aligning text-to-music evaluation with human preferences, arXiv preprint arXiv:2503.16669 (2025).

[11] H.-T. Hung, J. Ching, S. Doh, N. Kim, J. Nam, Y.-H. Yang, Emopia: A multi-modal pop piano dataset for emotion recognition and emotion-based music generation, arXiv preprint arXiv:2108.01374 (2021).

[12] L. N. Ferreira, J. Whitehead, Learning to generate music with sentiment, Proceedings of 20th International Conference on Music Information Retrieval (ISMIR) (2019) 318–325.

[13] J. Grekow, T. Dimitrova-Grekow, Monophonic music generation with a given emotion using conditional variational autoencoder, IEEE Access 9 (2021) 129088–129101.

[14] S. Sulun, M. E. Davies, P. Viana, Symbolic music generation conditioned on continuous-valued emotions, IEEE Access 10 (2022) 44617–44626.

[15] J. Huang, K. Chen, Y.-H. Yang, Emotion-driven piano music generation via two-stage disentanglement and functional representation, ISMIR (2024).

[16] D. Turnbull, L. Barrington, D. Torres, G. Lanckriet, Towards musical query-by-semantic-description using the cal500 data set, in: Proceedings of the 30th annual international ACM SIGIR conference on Research and development in information retrieval, 2007, pp. 439–446.

[17] S.-Y. Wang, J.-C. Wang, Y.-H. Yang, H.-M. Wang, Towards time-varying music auto-tagging based on cal500 expansion, in: 2014 IEEE International Conference on Multimedia and Expo (ICME), IEEE, 2014, pp. 1–6.

[18] S. L. J. J. T. K. J. N. Eunjin Choi, Yoonjin Chung, YM2413-MDB: A multi-instrumental FM video game music dataset with emotion annotations, in: Proc. Int. Society for Music Information Retrieval Conf., 2022.

[19] M. Soleymani, M. N. Caro, E. M. Schmidt, C.-Y. Sha, Y.-H. Yang, 1000 songs for emotional analysis of music, in: Proceedings of the 2nd ACM international workshop on Crowdsourcing for multimedia, 2013, pp. 1–6.

[20] A. Aljanaki, Y.-H. Yang, M. Soleymani, Developing a benchmark for emotional analysis of music,

PloS one 12 (2017) e0173392.
[21] J. A. Speck, E. M. Schmidt, B. G. Morton, Y. E. Kim, A comparative study of collaborative vs. traditional musical mood annotation., in: ISMIR, volume 104, 2011, pp. 549–554.
[22] H. Lee, E. Çelen, P. Harrison, M. Anglada-Tort, P. van Rijn, M. Park, M. Schönwiesner, N. Jacoby, Globalmood: A cross-cultural benchmark for music emotion recognition, arXiv preprint arXiv:2505.09539 (2025).
[23] X. Gao, D. K. Chen, Z. Gou, L. Ma, R. Liu, D. Zhao, J. Ham, Ai-driven music generation and emotion conversion, Affective and Pleasurable Design 123 (2024).
[24] J. A. Russell, A circumplex model of affect., Journal of personality and social psychology 39 (1980) 1161.
[25] M. Yik, J. A. Russell, J. H. Steiger, A 12-point circumplex structure of core affect., Emotion 11 (2011) 705.
[26] A. B. Warriner, V. Kuperman, M. Brysbaert, Norms of valence, arousal, and dominance for 13,915 english lemmas, Behavior research methods 45 (2013) 1191–1207.
[27] M. M. Bradley, P. J. Lang, Affective norms for English words (ANEW): Instruction manual and affective ratings, Technical Report, Technical report C-1, the center for research in psychophysiology ..., 1999.
[28] H. Liu, Y. Yuan, X. Liu, X. Mei, Q. Kong, Q. Tian, Y. Wang, W. Wang, Y. Wang, M. D. Plumbley, Audioldm 2: Learning holistic audio generation with self-supervised pretraining, IEEE/ACM Transactions on Audio, Speech, and Language Processing (2024).
[29] J. Copet, F. Kreuk, I. Gat, T. Remez, D. Kant, G. Synnaeve, Y. Adi, A. Défossez, Simple and controllable music generation, Advances in Neural Information Processing Systems 36 (2023) 47704–47720.
[30] J. Melechovsky, Z. Guo, D. Ghosal, N. Majumder, D. Herremans, S. Poria, Mustango: Toward controllable text-to-music generation, arXiv preprint arXiv:2311.08355 (2023).
[31] Z. Evans, J. D. Parker, C. Carr, Z. Zukowski, J. Taylor, J. Pons, Stable audio open, in: ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2025, pp. 1–5.
[32] Suno, Suno, https://suno.com/, 2024. Accessed: May-June, 2025.
[33] Udio, Udio, https://www.udio.com/, 2024. Accessed: May-June, 2025.
[34] J. Ho, A. Jain, P. Abbeel, Denoising diffusion probabilistic models, Advances in neural information processing systems 33 (2020) 6840–6851.
[35] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, Advances in neural information processing systems 30 (2017).
[36] H. W. Chung, L. Hou, S. Longpre, B. Zoph, Y. Tay, W. Fedus, Y. Li, X. Wang, M. Dehghani, S. Brahma, et al., Scaling instruction-finetuned language models, Journal of Machine Learning Research 25 (2024) 1–53.
[37] P. N. Juslin, From everyday emotions to aesthetic emotions: Towards a unified theory of musical emotions, Physics of life reviews 10 (2013) 235–266.
[38] X. He, N. Song, Emotional value in online education: A framework for service touchpoint assessment, Sustainability 15 (2023) 4772.

A. Details in Dataset Construction

We use the same prompt format for all generations. For a given emotion intent *word*, the text input to the TTM system was "*word*, instrumental". The reason we added 'instrumental' to the prompt was to ensure that only music clips without vocals would be generated, as vocal lyrics can influence the emotional expression of the song. For Suno and Udio, there is an 'instrumental' toggle option to prevent the models from generating vocal tracks. Thus, the text input contains only the emotion intent itself with no additional words, and the 'instrumental' option is enabled.

For open-source models, each clip was generated with a length of 30 seconds, after which a 10-second segment was randomly cropped from between the 15% and 85% marks of the full clip to exclude the intro and outro, which typically contain less information. It was impossible to control the exact length of the output generated by Suno and Udio, so we applied the same cropping procedure, with the only difference being that the full-length clips had variable lengths. The lengths of all generated clips were at least 30 seconds, which was sufficient to extract a 10-second segment.

As the sample rate of each model's generation was fixed per checkpoint, we could not unify the sample rates. The sampling frequency of each model is shown in Table 2.

Table 2
Information of TTM systems to generate music clips

Model	Checkpoint (Version)	Availability	Sample rate(kHz)
AudioLDM 2	audioldm2-music	open-source	16
MusicGen	musicgen-medium	open-source	32
Mustango	mustango	open-source	16
Stable Audio Open	stable-audio-open-1.0	open-source	44.1
Suno	Suno v4.5	commercial	44.1
Udio	Udio v1.5 Allegro	commercial	44.1

B. Details in Human Evaluation

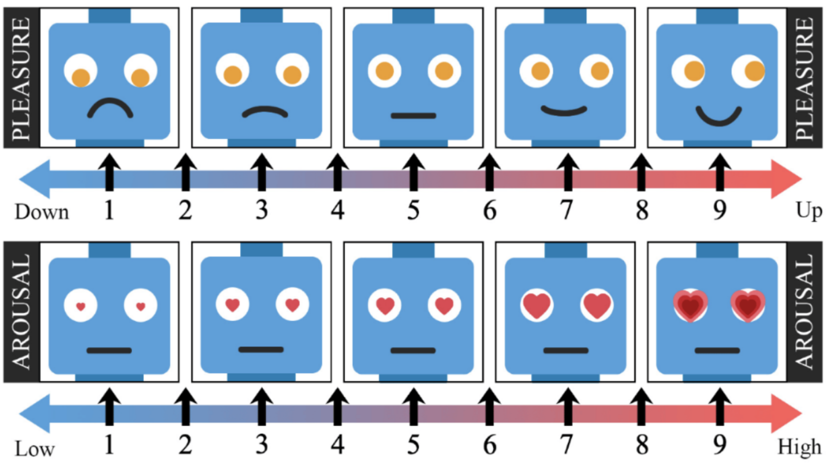


Figure 3: Survey images for valence (top) and arousal (bottom) rating questions. Adapted from He et al. [38].

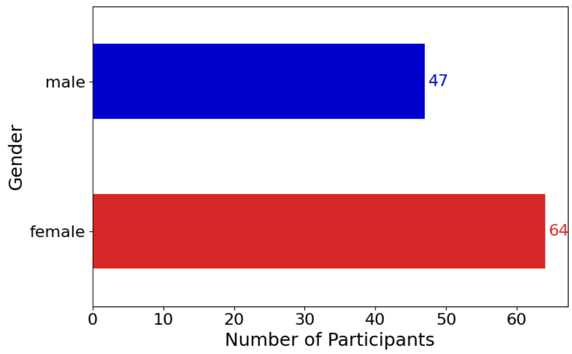
During the human evaluation conducted via the survey webpage, participants were presented with each clip one at a time and asked (in Korean): "Rate the pleasure conveyed by this music clip, with 1 meaning 'very unpleasant' and 9 meaning 'very pleasant,'" and "Rate the arousal conveyed by this music clip, with 1 meaning 'very relaxed' and 9 meaning 'very aroused.'" The images in Figure 3 were displayed alongside both questions. Participants selected integer scores from 1 to 9 for each question and were required to provide one rating per question to proceed to the next clip, similar to the static annotation method used in the DEAM dataset [20].

Before rating the clips, participants were asked to provide demographic information, such as gender and age. This procedure ensured that all participants were fluent Korean speakers aged at least 18. They were then shown a sample question and clip to confirm that they could hear the audio and understand the instructions. The sample clip was neither rated nor included in the results. The only other data collected was contact information for the purpose of providing a participation reward.

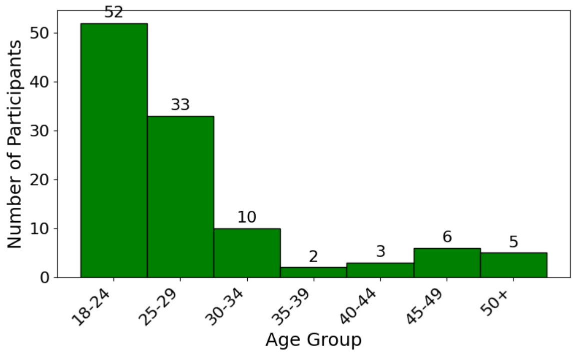
When selecting music clips for a new participant, priority was given to clips rated by fewer participants to help balance the number of ratings for each clip. However, it was difficult to completely balance the ratings, as some participants did not complete the survey. As a result, the number of participants who rated each clip ranged from 2 to 9, and we excluded 17 clips with three or fewer ratings, leaving 991 clips in the final dataset. Thus, the number of ratings on each clip ranges from 4 to 9.

C. Statistics on Participants and Number of Ratings

111 participants rated the generated music clips, resulting in a total of 6,162 ratings after excluding ratings on clips with three or fewer raters. For one participant, six ratings were removed because the participant was unable to listen to the last six samples due to a technical issue with the survey webpage. Detailed information about the participants population is shown in Figure 4, and number of ratings per clip is presented in Figure 5.



(a)



(b)

Figure 4: (a) Gender distribution of survey participants. (b) Age distribution survey participants.

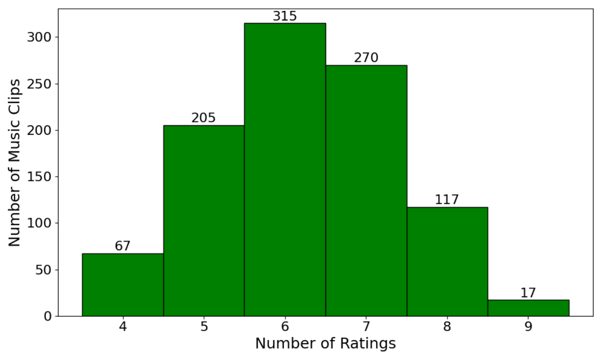
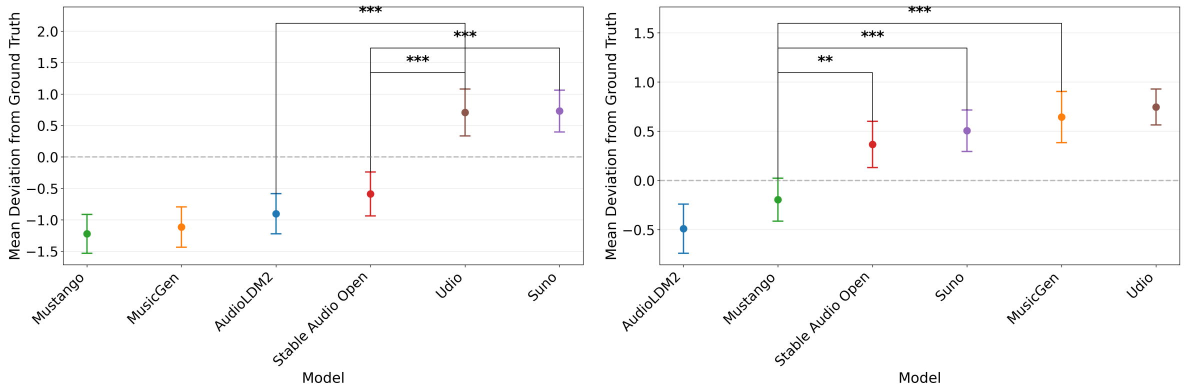


Figure 5: Distribution of music clips by number of ratings per clip.

D. Additional Results Figures and Tables



(a) (b)

Figure 6: Mean deviation plots for each model with 95% confidence intervals. Among all significantly different pairs, the three with the smallest differences are shown. (*: p<0.05, **: p<0.01, ***: p<0.001) (a) Mean valence deviations. (b) Mean arousal deviations.

Table 3
Pairwise comparison of mean valence and arousal deviations between TTM systems. Tukey's HSD test was used for multiple comparisons. p-values greater than 0.05 (not statistically significant) are shown in **bold**.

TTM systems			p-value for valence	p-value for arousal
AudioLDM 2	vs	MusicGen	**.9514**	.0000
AudioLDM 2	vs	Mustango	**.7693**	**.4636**
AudioLDM 2	vs	Stable Audio Open	**.7767**	.0000
AudioLDM 2	vs	Suno	.0000	.0000
AudioLDM 2	vs	Udio	.0000	.0000
MusicGen	vs	Mustango	**.9977**	.0000
MusicGen	vs	Stable Audio Open	**.2382**	**.5253**
MusicGen	vs	Suno	.0000	**.9582**
MusicGen	vs	Udio	.0000	**.9889**
Mustango	vs	Stable Audio Open	**.0862**	.0074
Mustango	vs	Suno	.0000	.0003
Mustango	vs	Udio	.0000	.0000
Stable Audio Open	vs	Suno	.0000	**.9565**
Stable Audio Open	vs	Udio	.0000	**.1780**
Suno	vs	Udio	**.9999**	**.6792**

Table 4
Pairwise comparison of mean absolute valence and arousal deviations across valence–arousal quadrants. Tukey's HSD test was used for multiple comparisons.

Quadrants			p-value for valence	p-value for arousal
Q1(high-valence, high-arousal)	vs	Q2(low-valence, high-arousal)	.0000	.0000
Q1(high-valence, high-arousal)	vs	Q3(low-valence, low-arousal)	.0000	.0000
Q1(high-valence, high-arousal)	vs	Q4(high-valence, low-arousal)	.9524	.0029
Q2(low-valence, high-arousal)	vs	Q3(low-valence, low-arousal)	.0146	.2310
Q2(low-valence, high-arousal)	vs	Q3(high-valence, low-arousal)	.0000	.7341
Q3(low-valence, low-arousal)	vs	Q4(high-valence, low-arousal)	.0000	.0187

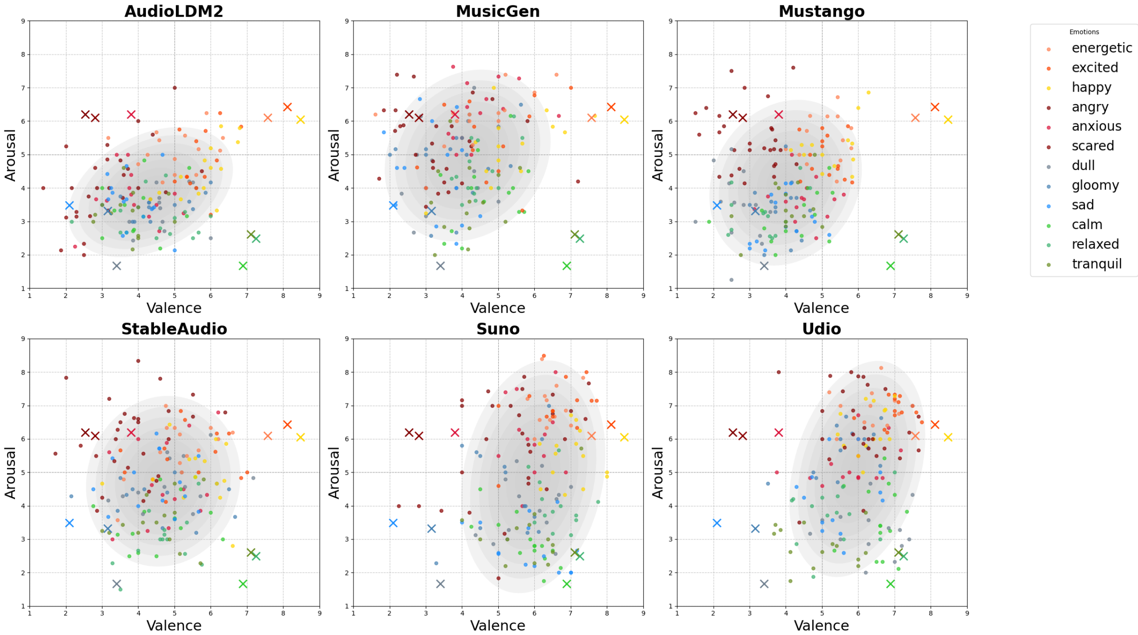


Figure 7: Scatter plot of average valence–arousal ratings per clip. Colors represent emotion intents; X-marks show ground truth; gray ellipses denote 95% confidence regions.

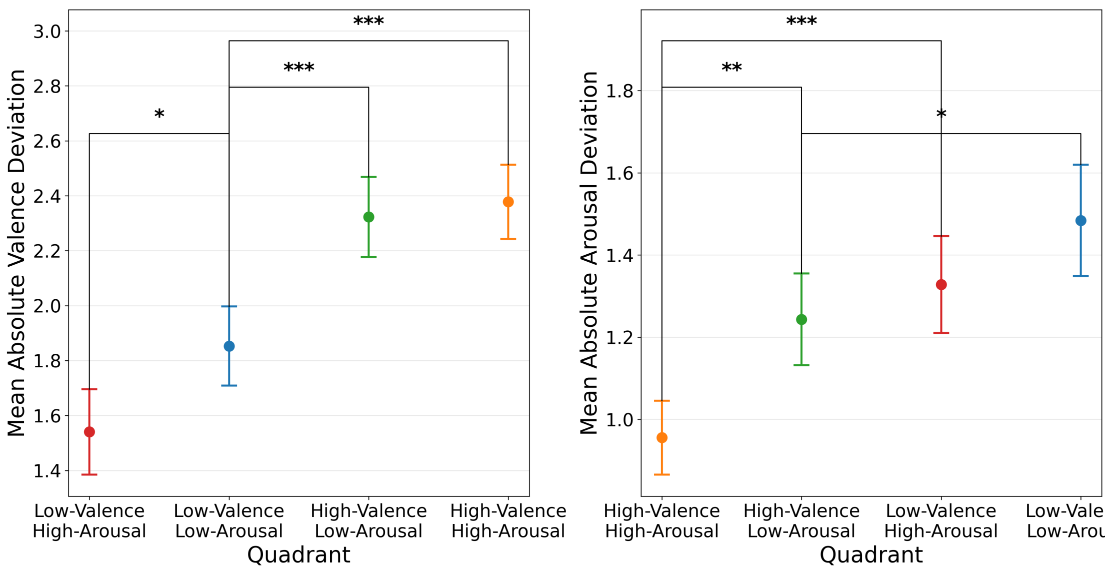


Figure 8: Mean absolute deviation plots for each valence-arousal quadrant with 95% confidence intervals. Among all significantly different pairs, the three with the smallest differences are shown. (*: p<0.05, **: p<0.01, ***: p<0.001) (a) Mean absolute valence deviations. (b) Mean absolute arousal deviations.

Table 5
Summary statistics of clip ratings by TTM system and valence-arousal quadrant.

TTM systems	quadrants	valence		arousal	
		mean	std	mean	std
AudioLDM 2	Total	4.34	1.16	3.87	0.99
	Q1	5.50	0.80	4.81	0.80
	Q2	3.39	1.02	3.96	1.05
	Q3	4.36	0.81	3.50	0.60
	Q4	4.10	0.89	3.23	0.67
MusicGen	Total	4.16	1.17	4.97	1.31
	Q1	4.86	1.24	5.63	1.04
	Q2	3.76	1.24	5.63	1.14
	Q3	3.58	0.94	4.67	1.24
	Q4	4.43	0.74	4.00	1.05
Mustango	Total	4.00	1.07	4.16	1.26
	Q1	5.16	0.58	5.22	0.70
	Q2	3.49	1.02	4.88	1.21
	Q3	3.50	0.83	3.28	0.87
	Q4	3.88	0.86	3.27	0.71
Stable Audio Open	Total	4.69	1.09	4.73	1.30
	Q1	5.33	0.99	5.38	0.88
	Q2	4.17	1.08	5.61	1.33
	Q3	4.67	1.10	4.58	0.74
	Q4	4.59	0.86	3.36	0.78
Suno	Total	5.97	1.02	4.87	1.78
	Q1	6.68	0.68	6.47	1.07
	Q2	5.50	1.16	5.87	1.44
	Q3	5.65	0.94	3.76	1.24
	Q4	6.05	0.82	3.36	0.91
Udio	Total	5.97	0.96	5.10	1.65
	Q1	6.52	0.63	6.65	0.63
	Q2	5.99	0.93	5.99	1.15
	Q3	5.81	0.92	4.41	1.31
	Q4	5.58	1.06	3.37	0.91